



NG80

HIGH THROUGHPUT

NANOMECHANICAL TESTING PLATFORM



NANOINDENTATION
Measure Hardness
and Elastic Modulus



SPM IMAGING
3D Topography and
Site-specific Analysis



HIGH SPEED INDENTATION
300x Faster Property
Mapping and Statistics

WHY CHOOSE NG80 ?

NG80 brings together multiple advanced technologies in one compact platform to deliver fast, accurate and repeatable nanomechanical testing for today's research challenges.



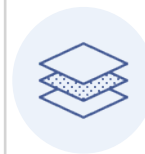
300× FASTER PROPERTY MAPPING

High Speed Indentation (HSI) enables rapid microstructural mapping and statistical analysis in a fraction of time.



NANOMETER PRECISION

Force resolution of 1 nN and displacement resolution of 0.006 nm ensure research grade accuracy.



MULTI-TECHNIQUE PLATFORM

Nanoindentation, SPM Imaging, Scanning Nanowear, HSI and High Temperature Testing in one integrated system.



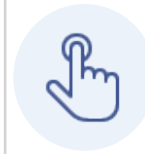
RELIABLE & REPEATABLE

Engineered for superior stability and low noise for consistent, repeatable results you can trust.



HIGH TEMPERATURE CAPABILITY

Optional high temperature stage enables nanoindentation at temperatures up to 600°C (as configured).



EASY TO USE SOFTWARE

Intuitive software with automated workflows simplifies testing, analysis and reporting.

IDEAL FOR



AUTOMOTIVE



AEROSPACE



BATTERY
MATERIALS



THIN FILMS &
COATINGS



SEMICONDUCTORS
& MEMS



BIOMATERIALS &
MEDICAL DEVICES



EDUCATION &
RESEARCH LABS

ADVANCED TECHNOLOGIES. POWERFUL INSIGHTS.

NG80 integrates multiple cutting-edge techniques to deliver comprehensive nanomechanical characterization with unmatched speed and accuracy.

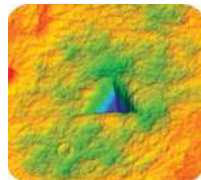


KEY TECHNOLOGIES

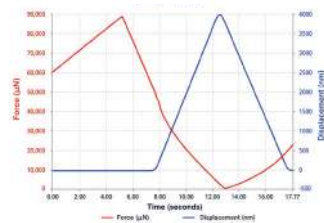


1. NANOINDENTATION

Measure Hardness and Elastic Modulus at the nanometer scale.

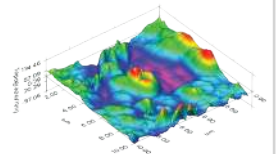


- Load and displacement control
- Accurate force and depth measurement
- Ideal for thin films, coatings and bulk materials
- Nano-scale mechanical characterization



2. IN-SITU SPM IMAGING

High resolution 3D topography for site-specific analysis and precise experiment targeting.

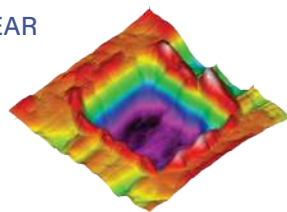


- 3D Topography with nanometer resolution
- Site-specific indentation with ± 10 nm accuracy
- Surface roughness and feature analysis
- Image size up to $50 \mu\text{m} \times 50 \mu\text{m}$

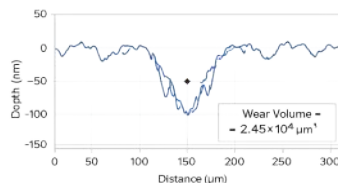


3. SCANNING NANOWEAR

Quantify wear behavior with sub-micron resolution and in-situ imaging.



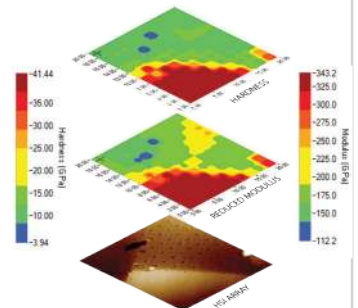
- Measure wear volume and wear rate
- Multiple pass wear tests at different normal forces
- Friction and wear mapping
- Real-time wear track analysis



4. HIGH SPEED INDENTATION

300× faster testing for rapid property mapping and statistical analysis.

- Up to 4 indents per second
- Large area property mapping
- Statistical distribution of mechanical properties
- 300× faster than conventional indentation



TECHNICAL SPECIFICATIONS



High Speed Indentation

High Speed Indentation: 4 indents per second



In-situ SPM Imaging

SPM Image Size: 50 μm $\times$ 50 μm

SPM Image Resolution: 256 $\times$ 256



Optical Microscope

Magnification: 10X (Infinity Corrected)
20X (Optional)

Optical Resolution: 1 μm

Working Distance: 34 mm

Illumination: Coaxial

Camera Resolution: 5 MP

Focus Mode: AutoFocus (Optional)



Positioning

X, Y, Z Stage Travel: 100 mm $\times$ 50 mm $\times$ 50 mm

XY Step Resolution: 50 nm

Z Step Resolution: 10 nm



Force

Noise Floor: < 200 nN

Resolution: 1 nN

Max. Normal Force: 10 mN



Displacement

Noise Floor: 4 indents per second

Resolution: 0.006 mm

Max. Normal Displacement: 5 μm



RESEARCH GRADE
PERFORMANCE



INTUITIVE
WORKFLOW



BUILT FOR
DURABILITY



GLOBAL
SUPPORT

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